

## BUA CHUM, Thailand

WMO: 484180

Lat: 15.267N

Lon: 101.187E

Elev: 51

StdP: 100.71

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 15.4       | 17.0 | 9.7                           | 7.5 | 20.2 | 11.4 | 8.5 | 21.0 | 8.4                   | 25.5 | 7.9 | 25.9 | 0.4                   | 50   | 0.334 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 10.7                   | 38.5            | 26.4 | 37.4 | 26.2 | 36.4 | 26.1 | 28.2                | 34.1 | 27.9 | 33.6 | 27.6 | 33.1 | 2.5                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.8                            | 22.6 | 30.8 | 26.5 | 22.2 | 30.5 | 26.2 | 21.8 | 30.1 | 91.3          | 34.4 | 89.5 | 33.6 | 88.2 | 33.4 | 30.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 6.6           | 5.7  | 4.8 | 13.0                       | 40.0 | 2.2                | 1.1 | 11.4                                               | 40.8 | 10.2       | 41.4 | 8.9        | 42.1 | 7.3        | 42.9 |     |     |
| (5)               |      |     | 11.1                       | 29.2 | 1.9                | 0.6 | 9.8                                                | 29.6 | 8.7        | 30.0 | 7.6        | 30.3 | 6.2        | 30.8 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 28.4   | 25.5  | 27.8  | 30.0  | 30.9  | 30.3  | 29.9  | 29.2  | 28.8  | 28.4  | 28.1  | 26.8  | 25.2  | (6)  | (7)  |
|                                                                        | DBStd       | 2.49   | 2.31  | 2.31  | 2.10  | 1.85  | 1.64  | 1.30  | 1.19  | 1.05  | 1.09  | 1.12  | 1.92  | 2.43  | (7)  | (8)  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  | (9)  |
|                                                                        | HDD18.3     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | (9)  | (10) |
|                                                                        | CDD10.0     | 6722   | 481   | 498   | 619   | 628   | 631   | 597   | 594   | 584   | 552   | 562   | 504   | 472   | (10) | (11) |
|                                                                        | CDD18.3     | 3682   | 223   | 264   | 361   | 378   | 372   | 347   | 336   | 326   | 302   | 304   | 254   | 214   | (11) | (12) |
|                                                                        | CDH23.3     | 41721  | 2108  | 2971  | 4632  | 4929  | 4592  | 4277  | 3896  | 3604  | 3174  | 3154  | 2448  | 1935  | (12) | (13) |
|                                                                        | CDH26.7     | 19654  | 925   | 1532  | 2574  | 2771  | 2326  | 2081  | 1731  | 1497  | 1202  | 1224  | 1009  | 784   | (13) | (14) |
| (14) Wind                                                              | WSAvg       | 1.1    | 1.5   | 1.4   | 1.4   | 1.1   | 0.9   | 1.0   | 1.0   | 0.8   | 0.5   | 0.8   | 1.2   | 1.6   | (14) | (15) |
| (15) Precipitation                                                     | PrecAvg     | 1143   | 10    | 8     | 48    | 82    | 141   | 131   | 127   | 182   | 260   | 126   | 21    | 6     | (15) | (16) |
|                                                                        | PrecMax     | 1477   | 49    | 60    | 142   | 196   | 231   | 235   | 261   | 293   | 403   | 362   | 160   | 65    | (16) | (17) |
|                                                                        | PrecMin     | 833    | 0     | 0     | 0     | 3     | 39    | 33    | 59    | 106   | 152   | 1     | 0     | 0     | (17) | (18) |
|                                                                        | PrecStd     | 161    | 16    | 12    | 38    | 44    | 46    | 53    | 47    | 49    | 68    | 75    | 32    | 14    | (18) | (19) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 35.3  | 37.8  | 39.5  | 40.1  | 39.1  | 37.7  | 36.4  | 35.2  | 34.6  | 34.6  | 35.3  | 34.8  | (19) | (20) |
|                                                                        |             | MCWB   | 24.0  | 25.2  | 25.4  | 26.7  | 27.6  | 27.1  | 27.2  | 27.1  | 27.7  | 27.0  | 25.8  | 24.9  | (20) | (21) |
|                                                                        | 2%          | DB     | 34.1  | 36.4  | 38.0  | 38.8  | 37.4  | 36.1  | 35.0  | 34.2  | 33.6  | 33.5  | 33.9  | 33.5  | (21) | (22) |
|                                                                        |             | MCWB   | 23.9  | 24.7  | 25.6  | 26.6  | 27.2  | 26.9  | 26.8  | 27.0  | 27.3  | 26.5  | 25.3  | 24.2  | (22) | (23) |
|                                                                        | 5%          | DB     | 33.0  | 35.3  | 37.0  | 37.5  | 35.9  | 34.9  | 33.9  | 33.3  | 32.7  | 32.7  | 32.8  | 32.2  | (23) | (24) |
|                                                                        |             | MCWB   | 23.6  | 24.5  | 25.6  | 26.5  | 27.0  | 26.9  | 26.7  | 26.9  | 27.0  | 26.1  | 24.9  | 23.6  | (24) | (25) |
|                                                                        | 10%         | DB     | 31.6  | 34.0  | 35.8  | 36.2  | 34.6  | 33.9  | 32.9  | 32.4  | 31.8  | 31.8  | 31.6  | 30.9  | (25) | (26) |
|                                                                        |             | MCWB   | 22.9  | 24.2  | 25.4  | 26.4  | 26.9  | 26.7  | 26.5  | 26.7  | 26.8  | 25.9  | 24.3  | 22.9  | (26) | (27) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 26.2  | 26.7  | 27.5  | 28.6  | 28.7  | 28.2  | 28.2  | 28.2  | 28.5  | 28.1  | 27.5  | 26.5  | (27) | (28) |
|                                                                        |             | MCDB   | 32.4  | 34.7  | 35.8  | 36.7  | 35.6  | 34.1  | 33.4  | 32.9  | 33.0  | 32.6  | 32.9  | 32.2  | (28) | (29) |
|                                                                        | 2%          | WB     | 25.3  | 26.0  | 26.9  | 27.8  | 28.1  | 27.7  | 27.6  | 27.7  | 27.8  | 27.5  | 26.8  | 25.5  | (29) | (30) |
|                                                                        |             | MCDB   | 31.5  | 33.9  | 34.8  | 35.5  | 34.7  | 33.5  | 32.8  | 32.4  | 32.1  | 31.9  | 31.7  | 31.1  | (30) | (31) |
|                                                                        | 5%          | WB     | 24.6  | 25.6  | 26.5  | 27.3  | 27.7  | 27.4  | 27.2  | 27.3  | 27.4  | 27.1  | 26.1  | 24.7  | (31) | (32) |
|                                                                        |             | MCDB   | 30.4  | 32.9  | 34.2  | 34.8  | 33.8  | 33.0  | 32.3  | 31.9  | 31.5  | 31.3  | 30.6  | 30.2  | (32) | (33) |
|                                                                        | 10%         | WB     | 23.9  | 25.0  | 26.0  | 26.9  | 27.2  | 27.1  | 26.8  | 27.0  | 27.0  | 26.6  | 25.5  | 23.9  | (33) | (34) |
|                                                                        |             | MCDB   | 29.3  | 31.8  | 33.6  | 34.2  | 33.0  | 32.5  | 31.7  | 31.5  | 31.1  | 30.4  | 29.4  | 29.3  | (34) | (35) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.3  | 12.6  | 11.6  | 10.7  | 9.1   | 8.5   | 8.0   | 7.8   | 7.2   | 7.7   | 9.6   | 11.3  | (35) | (36) |
|                                                                        |             | MCDBR  | 13.9  | 14.2  | 13.2  | 12.2  | 10.8  | 9.8   | 9.2   | 8.9   | 8.5   | 8.9   | 11.1  | 12.5  | (36) | (37) |
|                                                                        |             | MCWBR  | 5.1   | 4.5   | 3.6   | 3.4   | 3.1   | 2.8   | 2.8   | 3.0   | 3.2   | 3.1   | 3.7   | 4.5   | (37) | (38) |
|                                                                        | 5% WB       | MCDBR  | 11.7  | 12.6  | 11.8  | 11.1  | 9.7   | 8.9   | 8.4   | 7.9   | 8.1   | 8.1   | 9.4   | 11.5  | (38) | (39) |
|                                                                        |             | MCWBR  | 4.4   | 4.0   | 3.5   | 3.5   | 3.3   | 3.1   | 3.1   | 3.3   | 3.3   | 3.2   | 3.5   | 4.5   | (39) | (40) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.505 | 0.602 | 0.657 | 0.602 | 0.507 | 0.458 | 0.447 | 0.452 | 0.490 | 0.541 | 0.485 | 0.478 | (40) | (41) |
|                                                                        | taud        |        | 2.024 | 1.809 | 1.718 | 1.883 | 2.190 | 2.334 | 2.363 | 2.333 | 2.210 | 2.028 | 2.167 | 2.124 | (41) | (42) |
|                                                                        | Ebn at Noon |        | 785   | 727   | 701   | 741   | 803   | 834   | 846   | 850   | 821   | 767   | 800   | 798   | (42) | (43) |
|                                                                        | Edh at Noon |        | 167   | 216   | 243   | 206   | 149   | 128   | 124   | 130   | 146   | 171   | 144   | 148   | (43) | (44) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 4.85  | 5.26  | 5.45  | 5.87  | 5.60  | 5.29  | 5.01  | 4.87  | 4.71  | 4.83  | 4.89  | 4.78  | (44) | (45) |
|                                                                        | RadStd      |        | 0.32  | 0.21  | 0.30  | 0.25  | 0.32  | 0.29  | 0.27  | 0.29  | 0.26  | 0.30  | 0.24  | 0.17  | (45) | (46) |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (28 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                              |       | N/A     | N/A    | N/A     | N/A   | +0.26 | +0.28       | N/A     | N/A     | +100    |

Nomenclature: See separate page